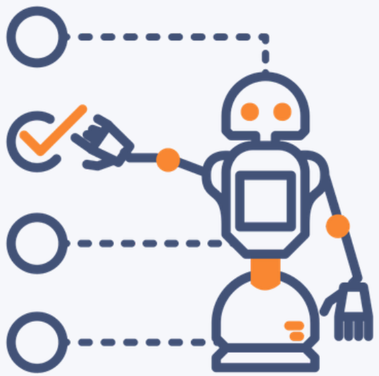
SC4063 PART 2 : DELIVERABLE 3

INTRODUCTION

- Agent-based architecture for automated PCAP analysis
- Multi-stage pipeline : ingestion, extraction, correlation, reasoning
- Integrates protocol analyzers (DNS, HTTP, TLS, beaconing, IOC)
- Produces structured outputs and campaign-level insights

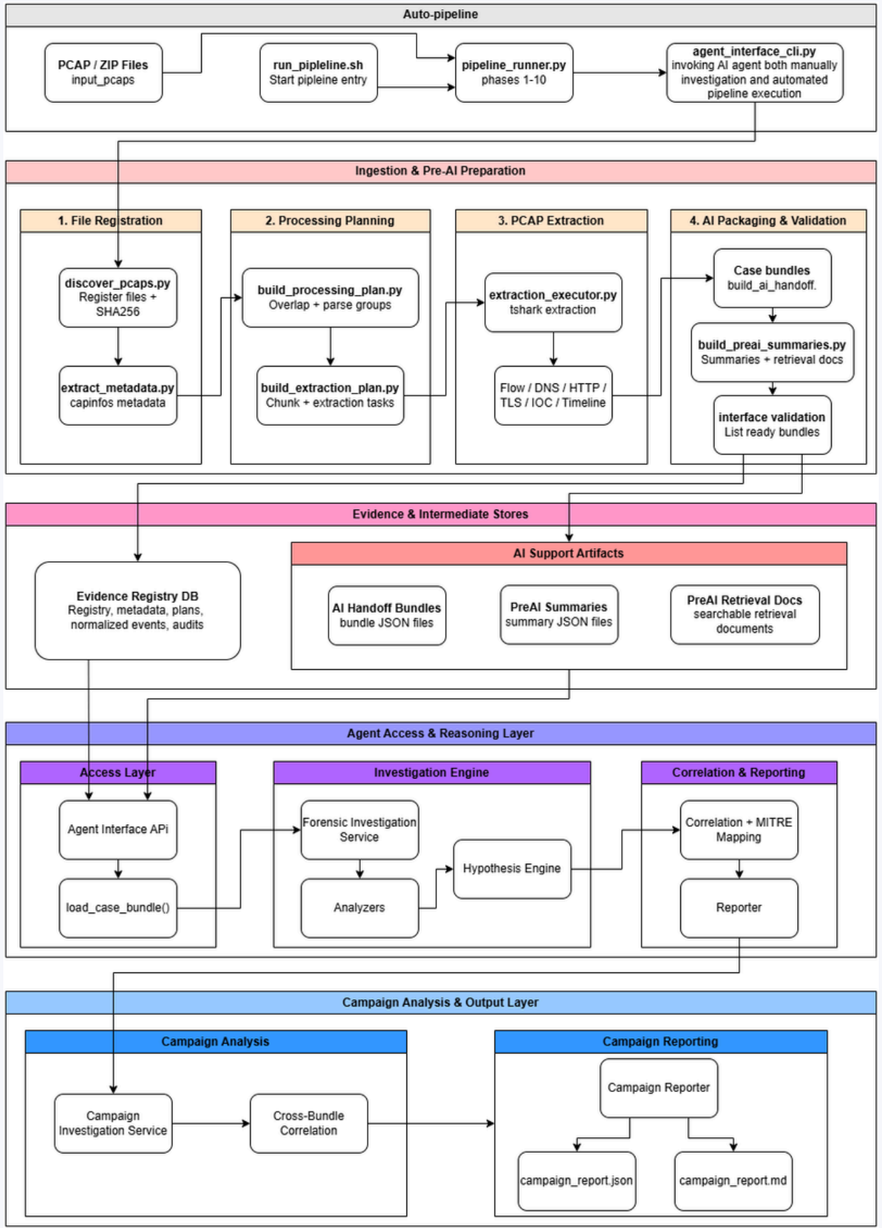


1. AGENT ARCHITECTURE

COMPONENTS + DATA FLOW

- PCAP files are ingested and preprocessed
- Ingestion pipeline extracts metadata and prepares data
- Agent acts as a central controller of the system
- Analysis modules extract protocol-level evidence
- Results are correlated and converted into insights

AGENT ARCHITECTURE IN-DETAIL



HIGH LEVEL TECH ARCHITECTURE

LAYER 1 · Web Interface	Flask · SSE · HTML/CSS/JS
LAYER 2 · Ingestion (P1–4)	Python · capinfos · SQLite
LAYER 3 · Extraction (P5)	tshark · Python multiprocessing (4 workers)
LAYER 4 · AI Reasoning (P6–10)	Ollama · Llama 3.1 · JSON Schema · MITRE ATT&CK
LAYER 5 · Correlation & Report	Mermaid · Markdown · JSON

TOOLS AVAILABLE TO THE AGENT

DNS Analyzer	Flags DGA-like domains, long / rare TLDs, NXDOMAIN bursts
HTTP Analyzer	Inspects headers, suspicious User-Agents, abnormal verbs / paths
TLS Analyzer	Missing SNI, self-signed certs, JA3 fingerprints
Beaconing Detector	Periodic callback intervals (C2 patterns)
IOC / Threat Intel	Known-bad IPs and domains enrichment
Ollama + Llama 3.1	Local LLM — narrative synthesis & MITRE mapping

DECISION-MAKING LOGIC

- Agent dynamically selects analysis modules
- Decisions are based on extracted evidence
- Results from one module guide the next step
- Correlation engine links related events
- Hypothesis engine generates possible attack scenarios

2. DEMONSTRATION

DEMO DATASET – LATEST RUN

Metric	Value
PCAP files processed	119
Total network events	4,898
Bundles formed	52
Date range	Dec 3, 2025 – Jan 29, 2026

Single end-to-end run analysed nearly 5,000 events across 119 PCAPs spanning ~2 months

GENERATED OUTPUTS

83

Findings

53 HIGH · 30 MEDIUM

10

Campaign findings

cross-bundle correlation

11

Flagged for review

human-in-the-loop

5

MITRE techniques

T1133 · T1078 · T1021 · T1071 · T1573

TOP ATTACKER

88.214.25.115 → 10.128.239.57:3389 (RDP brute-force, persistent)

Finding types: External Sensitive Access · Suspicious TLS Session · Potential Data Exfiltration · Suspicious DNS · Known Bad IP

SUSPICIOUS TLS SESSIONS – GROUNDING VERIFIED

Pattern Detected

- 39 unique external IPs connected to internal host 10.128.239.57 on port 3389 (RDP)
- All TLS sessions lacked SNI (hallmark of automated or malicious tooling)
- TLS also observed on non-stamdard ports alongside RDP
- Activity persisted from Dec 3, 2025 through Jan 29, 2026 (nearly 2 months)

Top Repeat External IPs

179.60.146.36	**20x**	**10.128.239.57: 3389**
194.165.17.11	19x	10.128.239.57: 3389
179.60.146.33	17x	10.128.239.57: 3389
141.98.11.114	12x	10.128.239.57: 3389
195.211.190.189	10x	10.128.239.57:3389

FINDINGS BREAKDOWN

External Sensitive Access

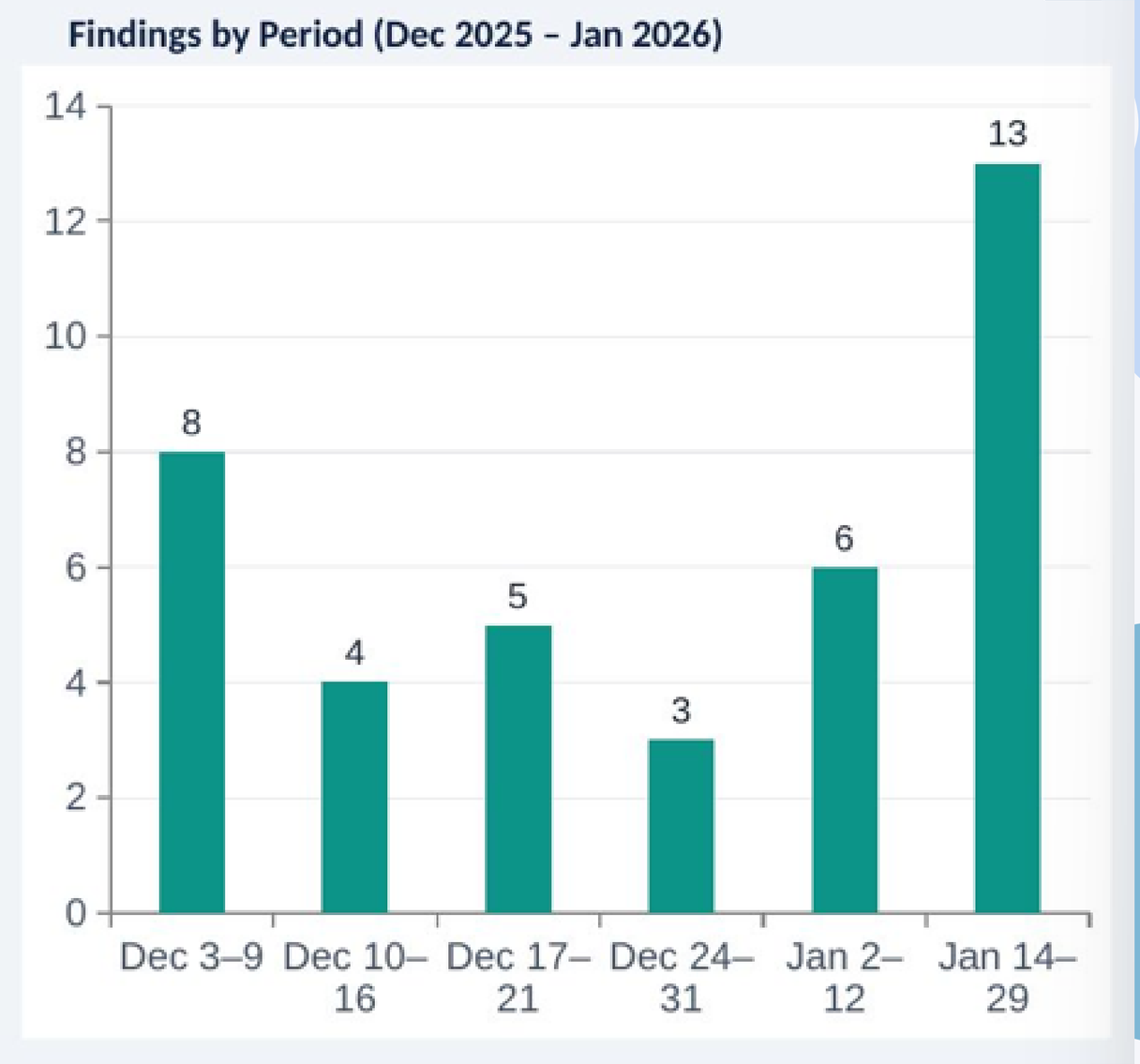
MITRE: T1133, T1078, T1021

Evidence items: 43

Suspicious TLS Sessions

MITRE: T1573, T1071

Evidence items: 30



3. KEY CHALLENGES

TECHNICAL CHALLENGES

- Large PCAP processing is slow and resource-heavy
- Correlation is hard across multiple extractor outputs (time, host, protocol)
- Uses normalized Zeek/Suricata-style events, but relies on fixed tools (capinfos, tshark)

FALSE POSITIVES / MISSED DETECTIONS

- Legitimate traffic (CDNs, updates, APIs) may appear suspicious
- Stealthy attacks can evade behavioural detection
- TLS encryption hides payloads - analysis limited to metadata (SNI, certs, JA3)

TOOL LIMITATIONS

- Extractors provide evidence, not conclusions
- Conclusions require combining multiple analyzers
- Detection depends on PCAP quality (missing / partial data reduces accuracy)

SCALING / PERFORMANCE ISSUES

- Large PCAPs increase processing time and storage
- Sequential pipeline adds latency
- Deeper analysis increases compute cost
- Trade-off : speed vs coverage vs confidence

4. GUARDRAILS AND SAFETY CONTROLS

Agentic AI systems can hallucinate - reporting IPs, domains, or ports that never existed in the data

Grounding check: every evidence is independently verified against indexed raw PCAP files

CAN WE TRUST AI FINDINGS?

Agent Finding

Finding: Suspicious TLS Session

Evidence: 10.128.239.57 → 88.214.25.115:34985

Confidence: 0.73 (MEDIUM)

MITRE: T1573, T1071

Does it match?

Raw PCAP Events

tls event

src: 10.128.239.57
dst: 88.214.25.115
dst_port: 34985
sni: (null)

flow event

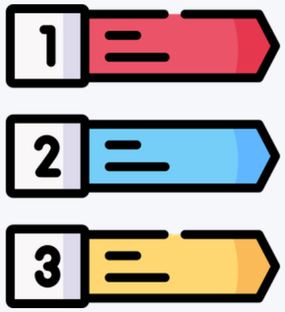
10.128.239.57:3389 → 88.214.25.115:34985

AGENT CONSTRAINTS

- Fixed investigation workflow
  - (Summary → DNS/HTTP/TLS → IOC → MITRE → Findings)
- Uses **AgentInterface + predefined analyzers only**
- No direct DB queries or raw PCAP parsing
- External tools restricted (e.g., tashark, capinfos)

VALIDATION METHODOLOGY

Index Raw Events

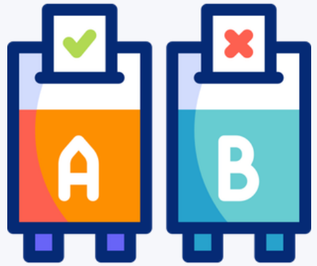


- Parse each bundle's ai_handoff_bundles JSON
- Build lookup sets: DNS domains, TLS (src, dst, port) tuples, flow pairs and all IP addresse

Check Each Evidence Item

For every evidence item in every finding, run source-specific checks (DNS, TLS, beaconing against the index)

Record Pass/Fail



- Each item is marked Grounded or Ungrounded.
- Failures log the exact reason for audit purposes.

HUMAN-IN-THE-LOOP

- High-risk / low-confidence → **analyst review**
- Reports include:
  - Confidence level
  - Evidence items
  - Risk (FP /FN)
  - Limitations
- Agent = **Decision support**, not replacement

HUMAN-IN-THE-LOOP

- 11 of 83 findings auto-flagged for analyst review
- Reports include: confidence level, evidence items, FP/FN risk, limitations
- Agent = decision support, not replacement for the analyst
- Conservative defaults: low-confidence findings are downgraded, never auto-actioned

11

findings flagged for human review

from 83 total — 13.3 %

5. COST AND EFFICIENCY

COMPUTATION OF COST PER RUN (ESTIMATED OR MEASURED)

- Local execution (CPU-based)
- Assumed rate: $0.05 per CPU-hour
- Total runtime: 94.1 minutes
- Estimated cost: ~ $0.078 per run

Key Points

- No cloud LLM cost (local pipeline)
- Cost mainly comes from CPU processing of PCAPs

- Note: All runtime and cost values in this table are based on the saved pipeline_run_metrics.json file from a successful full pipeline run. Additional task-level efficiency statistics are derived from evidence_registry.db using injestion/view_run_metrics.py.

RUNNING COST — BASELINE vs OPTIMISED

Metric	Baseline (1 worker)	Optimised (4 workers)	Improvement
Runtime	10.06 hrs	1.57 hrs	**6.4× faster**
Total elapsed	36,216 s	5,645 s	**−84 %**
Phase 5 workers	1	4	**4× parallel**
Findings produced	46	83	**+80 %**
Est. compute cost*	~$0.50	~$0.078	**84 % cheaper**

** Estimated at $0.05/hr for local compute (electricity + machine amortisation)*

TIME TO COMPLETE ANALYSIS

Stage	Runtime	Notes
Full Pipeline	94.1 min	End-to-end
Evidence Extraction	93.9 min	Main bottleneck
Agent Investigation	0.49 sec	Near real-time
Campaign Analysis	0.46 sec	Very fast
Final Synthesis	0.06 sec	Negligible

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TRADE-OFFS — SPEED · ACCURACY · COST

SPEED

4 parallel workers in Phase 5 (configurable in the web UI). Scales to 8 / 16 workers on more powerful hardware.

ACCURACY

Rule-based detection + LLM verification. 83 findings with 100 % grounding in evidence; 11 flagged for review.

COST

Local LLM (Ollama + Llama 3.1) → zero marginal cost per run. No cloud API, no vendor lock-in.

6. CONCLUSION

CONCLUSION

- End-to-end agent-driven PCAP analysis pipeline
- Multi-analyzer correlation (DNS, HTTP, TLS, beaconing) with AI reasoning
- Evidence-based, explainable outputs (no hallucination, traceable findings)
- Practical guardrails: confidence scoring + human-in-the-loop
- Limitations: data quality, encrypted traffic (TLS), cost–speed trade-offs
- Towards scalable and automated network forensics

THANK YOU